

Chapter 6

Understanding Money

Portions of this chapter have been adapted, rewritten, or expanded from the following works: The Bullish Case for Bitcoin by Vijay Boyapati (2021), The Bitcoin Standard by Saifedean Ammous (2018), and Layered Money by Nik Bhatia (2021). The author is grateful for their foundational research and explanation, which have propelled the author's understanding of money and Bitcoin and have helped inform the ideas presented in the chapter by piecing together and further expounding on their writings.

Understanding Bitcoin requires asking a simple question that turns out to be a very complicated one to answer. That simple question is “What is money”? In essence, Bitcoin is simple: it is a medium that holds value in a trustless and permissionless way across time and space, which no other party can steal, dilute, inflate, control, confiscate, expropriate, or coerce. But how do we arrive at this conclusion? What even is a dollar? And why do groups of unelected economists get to decree that some pieces of papers are more valuable than others? There are many questions yet to be answered.

Throughout human history, many things have served the function of money. Most notably being gold and silver, but also copper, seashells, glass beads, large stones, salt, cattle, government paper, and even alcohol, toilet paper, and cigarettes in particular cases and times. People's choices are subjective, and there is no definitively right or wrong choice to money. But there are, however, latent consequences behind people's choice of money throughout civilisations. Most people who live to average life expectancy spend roughly 90,000 hours in their lifetimes working for money (4), but unfortunately, very few take the time to understand what money really is.

Gensis

When God created the world and all its fullness, He welcomed humanity to a world of abundance and all-sufficiency, to live a life trusting and depending on Him as His universal counterpart. But man fell, we decided to sever our dependence on God, and rule the world according to our knowledge and judgment of good and evil. And in the process of man's fall, trust was lost between man and God (Gen. 3:8) and between man and man (Gen. 3:12, 4:8), and the abundance of the earth was taken away (Gen. 3:17).

In a world of scarcity and a lack of trust, to survive we needed to toil, we needed to trade, but we have no trust in each other. Exchanging the goods and services as the fruits of our labour face to face and good for good through bartering was only feasible in small communities, at few places, at few times, for few goods. As mankind filled the earth and expanded our population and

communities, we needed a trusted medium of exchange to trade the fruits of our labour for the fruits of others' labour in more places, at more times, for more goods. Man needed something that stored and preserved value in consensus among its users to be a medium of exchange, so that he could work, trade, eat, live, and multiply. This is how the concept of money came into the world: a medium of exchange for value.

Before elucidating the intricacies of what makes money “money”, great emphasis should be put on differentiating the concept of money from the underlying object used as money. The Lord told us that it is the love of money that is evil (1 Tim. 6:10), and the concept of money that will be abolished in the coming age (Luke 16:11), but things used as money in this age like gold, silver, precious minerals, and paper will still exist into the next age after the Lord returns.

Let's consider gold as one such underlying object used as money. In a story about a greedy king told by the Greeks, King Midas foolishly asked that everything he touched turn to gold, and to him it was granted. At first, he rejoiced as objects transformed into gold, making him unimaginably wealthy. But soon, he realised this curse, his food, drink, and even his beloved daughter turned to gold at his touch. Starving and grief-stricken, he begged that his wish could be reversed.

Hence, we see that money is not the underlying object, but rather its monetary value (or purchasing power in economic terms), agreed upon from the consensus of its users. Therefore, in reality, nobody wants only gold; instead, we want the value gold is able to hold and store, to be used for future transactions. As told in the story of Midas, if one were only to have gold and nothing else, that person would become very miserable very quickly.

Economic Energy

Key words

- **Types/Forms of money:** *Different things that humans consider to be money, e.g. gold, silver, fiat currency, and bitcoin.*
- **Fiat currency:** *Money issued by a government, whose value and supply is controlled by a central authority, commonly referred to as fiat.*
- **Monetary value:** *The value of goods and services that a type of money can be exchanged for; frequently used as the aggregate value of all of one type of money.*
- **Purchasing power:** *Interchangeable with monetary value; frequently used as the monetary value of one type of money on a unit basis, e.g. per \$100.*

Money (not the underlying object, but the value it holds) is the representation of the time and energy required to produce goods and services. It can also be viewed as value, as stored economic energy. Since the formation of early human communities to modern civilisations, we

work for money; hence, your money is the stored units of the time and energy you expended and worked for.

$$\text{Money you earned} = \text{Your expended time} + \text{Your expended energy}$$

Thinking of money's value as equivalent economic energy, we can transform it into chemical potential energy as food for our body, convert it into heat and electrical energy to heat and power our homes, and exchange it for kinetic energy as we fuel our cars or hop on aeroplanes. The more economic energy we possess, the greater the quantity of other forms of energy we can convert into and utilise. With more money, I can buy more steaks for dinner, or afford to travel by plane instead of walking on foot to another city. Naturally, being a form of energy, it too shall then obey the laws of thermodynamics established at the creation of the universe (Job 38:33).

The First Law – Energy cannot be created or destroyed, only transferred.

Man himself cannot create nor destroy energy; this law also holds true for economic energy, as the total aggregate monetary value in an economy. If a central bank decides to print money (in this case, fiat money) and double the supply of money in an economy overnight whilst the goods and services of that economy remain constant, the purchasing power of that fiat money is expected to halve (or the price of those goods and services to double). Aggregate monetary value, hence, the total economic energy within the economy, remains unchanged.

Another example, if I were to wave a magical wand and half of the gold in the world vanishes, we can expect the price of gold to double immediately. Again, the total monetary value of gold remains unchanged, the total economic energy of the aggregated gold supply remains constant.

Hence, the first law of the conservation of energy elucidates that “free money” from printing money by banks out of thin air isn't free, because energy cannot be created from nothing; the purchasing power of every holder of that currency must decrease proportionally to the amount of new money created. By printing money, the system permanently and covertly transfers a portion of the economic energy from everyone else's current holdings of that money into the newly created supply, being the most significant (but not the only) cause of the loss of purchasing power of fiat currencies, or inflation as commonly known.

The Second Law – The total entropy of a system always increases over time. Energy transformations are never 100% efficient.

Due to entropy, there is always inefficiency, dispersion, and degradation of every form of energy over a long enough period of time. Monetary value will always tend to zero, as the underlying object viewed as money degrades over time due to entropy. The copper coin will rust, the silver will lose its shine, the paper will decompose, and the plastic will fade.

Without accounting for the cost a central entity imposes on centralised fiat currency through inflation and taxation, the dispersion of economic energy will continuously incur costs from the maintenance of the underlying object used as money (storage, security, management fees). This elucidates that no one form of money can maintain 100% of its value forever (James 5:3), with the costs of entropy defining the useful life of each type of money.

The Third Law – As the energy of a system approaches absolute zero (non-existence), the entropy approaches a constant minimum.

The third law of thermodynamics shows it's impossible to achieve a state of perfect economic efficiency (zero waste, zero friction, zero inefficiency, no entropy) with money as stored economic energy. As long as money is continued to be used, a state of maximum efficiency in an economy can never be achieved.

Therefore, achieving a maximally efficient economy (entropy approaches a minimum) requires the complete abolition of money (energy approaches absolute zero). Hence, we now see why money was not in God's perfect will, and the concept of money emerged wholly through man; the concept of money for the transaction of monetary value is inefficient in a perfect world.

This fascinatingly leads us back to the world of abundance that God had created at the end of the sixth day of creation in Genesis chapter two, and the world to come in the next age as the restoration and enlargement of the Garden of Eden (Rev. 22:1-2), where man depends on God and fully trusts in each other, there is no money, only abundance and sufficiency. A world where he who has much had no excess, and he who gathered little had no lack (2 Cor. 8:15). A world where we do not need to account for the monetary value of transactions, because we can fully trust and rely on each other as God's children. A world where we always give out and receive freely from each other, because we know that God loves a cheerful giver (2 Cor. 9:7), and He Himself will always provide.

The Evolution of Money

Key words

- ***Monetary properties:*** *Various specific unique properties attributable to the underlying object used as money; the strengths and weaknesses of each property influence the pace of its monetisation, and its quality to function as money.*

There is an obsession in modern monetary economics with the medium of exchange role of money. In the 20th century, states have monopolised the issuance of money through central banking and continually undermined its use as a store of value, fostering a false belief that money is only defined as a medium of exchange. The steps of a money's monetisation and dual identity as a store of value and medium of exchange is analogous to the construction of a skyscraper, with

a good's store of value capability being the skyscraper's foundation and medium of exchange use cases being its stories.

To build the skyscraper upwards, it must have a firm foundation downwards. Without a firm foundation, the skyscraper would be insecure and unsafe, rendering additional stories useless and hazardous to utilise, regardless of how well designed, decorated, or furnished they are. On the other hand, without constructing any additional stories upon the foundation, the skyscraper would be very unspectacular and not function to its full potential. Hence, both the foundation and the stories of the skyscraper are crucial for its existence and function. When people view the skyscraper, their attention is drawn to its iconic architecture and vibrant night lights. However, for its architecture and functions of hosting offices, hotels, recreation centres and more to be possible, it unquestionably must first have a firm foundation into the earth.

The emphasis on money's characteristics of being a medium of exchange without considering the properties that enable it to store value, is likened to building upwards without laying a foundation. Without any value and the durability thereof, any use cases built on top is void. Money has always evolved in stages, with the store of value role preceding the medium of exchange role, being a foundation for its function. (1)

The 4-Stage Evolution of Money

1. **Collectible:** In the very first stage of its evolution, money will be demanded solely based on its peculiar properties, such as its scarcity and shininess. Shells, beads, and gold were all collectibles before later transitioning to the more familiar roles of money.
 - Notable monetary properties: Scarce, durable
2. **Store of value:** The ability to hold value over time. Once it is demanded by enough people for its peculiarities, money will be recognised as a means of storing and preserving value over time. As a good becomes more widely recognised as a suitable store of value from consensus, the value it holds will be maintained, and even appreciate, as more people demand it for this purpose.
 - Notable monetary properties: Scarce, durable, portable, established history, censorship resistant
3. **Medium of exchange:** The ability to transfer value across space. When money is fully established as a store of value, its purchasing power will stabilise. Having stabilised in purchasing power, the opportunity cost of using money to complete trades will diminish to a level suitable for use as a medium of exchange. Unique monetary properties will allow money to transfer value across space. Once at this stage, the underlying object will commonly be known as "money".

- Notable monetary properties: Portable, divisible, verifiable, established history
4. **Unit of account:** When money is widely used as a medium of exchange, goods will be priced in terms of it. For example, the exchange ratio against money will be available and fixed for most goods. For example, \$5 for a cup of coffee today, tomorrow, and next week, not fluctuating like \$4.96 today and \$5.03 tomorrow.
- Notable monetary properties: Portable, divisible, verifiable, fungible, established history

A type of money that is not yet a unit of account may be thought of as being “partly monetised”. Today gold fills such a role, serving as a store of value but having been stripped of its medium of exchange and unit of account roles by government intervention. It is also possible that one type of money fills the role of medium of exchange while another fills the other roles. This is typically true in countries with weaker economies, such as Argentina or Zimbabwe, where the local fiat currency is used for daily transactions across space, but no one will want to store their savings in their local currency for storing value over time.

In the United States of America, the dollar seamlessly serves the three functions of money: providing a medium of exchange, a unit for measuring the cost of goods, and an asset where value can be stored. In Argentina, on the other hand, while the peso was used as a medium of exchange for daily purchases of goods, no one used it as a store of value. Keeping savings in the peso was equivalent to throwing away money. So people exchanged any pesos they wanted to save for dollars, which kept their value better than the peso. Because the peso was so volatile, people usually remembered prices in dollars, which provided a more reliable unit of measure over time.

Bitcoin is currently transitioning from the second stage (store of value) of monetisation to the third stage (medium of exchange). It will likely take several years for Bitcoin to transition from an incipient store of value to a recognised medium of exchange. It is striking to note that the same transition took many centuries for gold. No one alive has seen the real-time monetisation of a good as is taking place with Bitcoin at such an accelerated pace. (1)

This now begs the question, what attributes of the underlying object make humans want to attribute value to it? And enable it to gradually evolve in the four-stage monetisation process?

Monetary Properties

When stores of value compete with each other, it is the specific properties of the good that allow it to outcompete another good at the margin and increase demand for it over time. While many goods have been used as stores of value or “proto-money”, certain properties emerged that were particularly demanded, allowing moneys with these attributes to outcompete other moneys and evolve through the four-stage monetisation process. Such properties are:

- **Durability:** The good must not be perishable or easily destroyed. Thus, wheat or 4-leaf clover is not an ideal store of value.
- **Portability:** The good must be easy to transport and store, making it possible to secure it against loss or theft and allowing it to facilitate long-distance trade, minimising transaction costs. A cow is thus less ideal than a gold bracelet.
- **Fungibility:** One specimen of the good should be interchangeable with another of equal quantity. For example, gold can be cut, melted, moulded, and formed into various shapes and sizes, but its properties remain the same, because the gold atoms remain the same. Thus, gold is better than diamonds, which are irregular in shape and quality.
- **Verifiability:** The good must be easy to quickly identify and verify as authentic. Easy verification increases the confidence of its recipient in trade and increases the likelihood that a trade will be completed.
- **Divisibility:** The good must be easy to subdivide. While this attribute was less important in early societies where trade was infrequent, it became more important with the exponential growth in the frequency and velocity of trade.
- **Scarcity:** The good must not be abundant or easy to either obtain or produce in quantity. Scarcity is perhaps the most important attribute of a store of value, as it taps into the innate human desire to collect that which is rare. It is the primary source of the original value of the store of value.
- **Established history** (trust & adoption): The longer the good is perceived to have been valuable by society, the greater its appeal as a store of value. A long-established store of value will be hard to displace by a new upstart money.
- **Censorship-resistance:** A new attribute of money, which has become increasingly important in our modern, digital society with pervasive surveillance and monopolised control of money, is censorship-resistance. That is, how difficult is it for another party, such as a bank or a state, to prevent the owner of the good from keeping and using it. Any type of money, regardless of how excellent the previous attributes are, is worthless if the government simply decides it is no longer yours, a critical attribute as discussed in the in earlier chapters.

Before beginning this chapter, have you ever wondered, why did gold come out on top in its monetisation against diamonds? If my teeth are very scarce, why aren't they worth millions each? Just some food for thought. Well, by accessing the monetary properties of things, we can determine if our candidate will become an ideal money.

Knowing the monetary properties of underlying objects really illustrates why some things are money and why some are not, why gold outcompeted diamonds, why the average lifespan of fiat currencies beginning since the 20th century was only 27 years, (5) and why my teeth will never be considered as money despite their rarity.

Monetary Properties of Monies the 21 st century					
	Bitcoin	Gold	Diamond	Fiat Money	My Teeth
Durable	A+	A+	A	C	B
Portable	A+	C	B	A	B
Fungible	A+	A	F	A	F
Verifiable	A+	C	C	A	F
Divisible	A+	A	F	A	F
Scarce	A+	A+	A	F	A+
Established history	C	A+	A	B	F
Censorship resistant	A+	C	C	F	F

In a very simple way, monetary properties of things give rise to various types of money, and the quality of monetary properties each type of money possesses determines the ability to advance and the pace of advancement of that money's four-stage monetisation, from a collectible to an unit of account. Monetary properties of things also gauge that type of money's ability to store and transfer economic energy (in other words, value). A good with better monetary properties tends to better store and transfer economic energy, thus people tend to park more of their economic energy in that good, further accelerating its monetisation, growth, and adoption.

	Key Monetary Properties
Store of value	Durability, verifiability, scarcity, established history, censorship resistance
Transfer of value	Portability, divisibility, fungibility, verifiability

Monetary Properties of Bitcoin

Now we have arrived at the understanding that:

- Money is not the underlying object itself, but is its monetary value that we attribute to it; the underlying object often has little to no utility value.
- Money functions as a store of value and a medium for the storage and transfer of economic energy across space and time.
- Money is defined by the monetary properties of the underlying object.
- Monies typically evolve through four stages of monetisation, and can devolve (demonetise) when better forms of money emerge.

Now we can answer the question, “What is Bitcoin”? In the simplest way, Bitcoin is a type of money, and its monetary properties rate it subjectively as the apex money of the 21st century. In fact, the question “what is Bitcoin” is equally valid as “what is gold”. Digging into the finest details of gold’s atomic structure, electron configuration, and electronegativity of this yellow metal can never explain why gold has value as the oldest and most recognised money in the world. However, by understanding the concept of money and accessing its monetary properties, we will be able to determine whether any good can function as money and assess its capacity as a medium for storing and transferring economic energy.

It sounds too good to be true that Bitcoin achieves an almost perfect score across all monetary properties. Let us objectively and more closely examine the attributes of Bitcoin that define its monetary properties.

Durability: Bitcoin, being the first truly intangible, neutral, and decentralised money in the world, does not suffer from the entropic degradation that all tangible moneys have. It is unaffected by the problems physical money have, like rust, maintenance, obsolescence, accident, theft, or natural disasters; it is unaffected by the problems digital currencies have, like inflation, taxation, and politics; it is unaffected by the problems of physical assets, like rates tax, rent control, law, tenants, leaky roof, insurance, crime, decay, regulation, and more. All of the listed items and more, are elements of entropy in the second law of thermodynamics that naturally erode the economic energy stored within goods, reducing their durability.

Bitcoin, on the other hand, represents a digital transformation of money as pure economic energy; its intangible nature makes it immune to the entropy all physical moneys (or for that matter, all physical things) suffer from. And itself requires no cost of maintenance to possess or carry.

Portability: Bitcoin is the most portable store of value ever used by man. Private keys representing any value of bitcoins, from a single cent to millions of dollars, can be stored on a tiny USB device and easily carried anywhere. Furthermore, equally valuable sums can be transmitted between people on opposite ends of the earth near instantly with final settlement. And finally, once you commit your private key to memory, you now carry that amount of bitcoins in your brain wherever you go, known commonly as “brain wallet”. In a sense, Bitcoin enables our brain to be

our bank; it is a money that empowers everyone to be their own bank, and by nature, a money available to everyone by birth.

Fungibility: As physical money, gold provides the standard for fungibility. When melted down, an ounce of gold is indistinguishable from any other ounce. Bitcoins are fungible in the network, meaning that every bitcoin, and every fraction of a bitcoin of the same quantity, is of also of the same quality and value on the Bitcoin network.

Verifiability: In the 21st century, contrary to the findings of the alchemists, gold is no longer immune to being counterfeited. Sophisticated criminals have used gold-plated tungsten as a way of fooling gold investors into paying for fake gold. Bitcoins, on the other hand, can be verified with mathematical certainty. Using cryptography, the owner of a bitcoin can publicly prove they own the bitcoin on the blockchain. It is simply impossible to forge bitcoins on the blockchain given the rapid growth of the network and the energy used to secure the network. The Bitcoin network itself is updated and audited every ten minutes on average by tens of thousands of globally decentralised nodes (run by voluntary node operators) to settle current transactions and confirm all past transactions of the network.

Divisibility: One bitcoin can be divided down to a hundred millionths ($1/100000000$) of a unit, known as one satoshi (or Sat), as the smallest base unit, and transmitted at such infinitesimal amounts. Fiat currencies are typically divisible down to pocket change, which has little purchasing power, making fiat currencies divisible enough in practice. Gold, while physically divisible, is immensely energy-consuming and thus highly impractical for day-to-day transactions.

Scarcity: The attribute that most clearly distinguishes Bitcoin from other monies is its predetermined absolute scarcity. By design, at most, 21 million bitcoins can ever be created. This gives us a known percentage of the total possible supply when one holds bitcoins. For instance, an owner of 21 bitcoins would know with absolute certainty that they hold 0.0001% of the

Gold production, 1815 to 2015

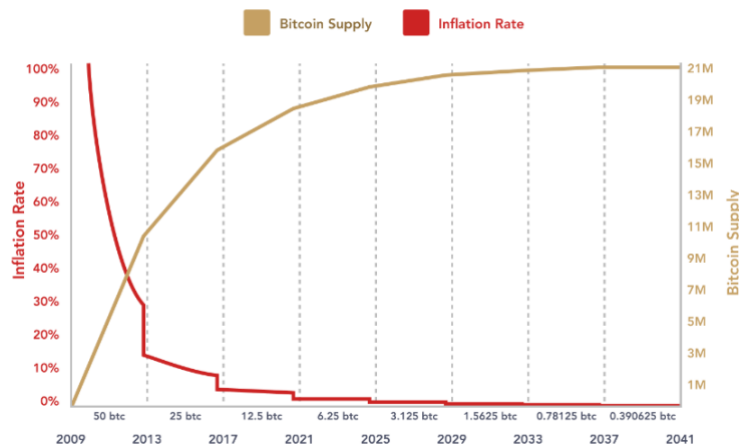
Annual gold production, measured in metric tons per year.



Data source: Clio Infra & USGS

OurWorldinData.org/fossil-fuels | CC BY

Bitcoin Issuance Schedule



aggregate supply. The same cannot be said of fiat currencies or precious metals, where their supply is scarce relatively, and the limit to their aggregate supply is still unknown.

Despite being digital, Bitcoin is designed to provide absolute scarcity, which is why it has the potential to be such an effective form of money. There will only ever be 21 million bitcoins, and 21 million is a scarily small number in relative and absolute terms. The Federal Reserve (central bank of the U.S.) created \$100 billion dollars within just a week* (4762 times the figure of dollars to Bitcoin in one week), with the click of a button. To provide a broader context, the Federal Reserve, the Bank of Japan, and the European Central Bank have collectively created north of \$17 trillion (\$17,000,000,000,000) dollars in new money in the decade from 2008 to 2018 alone. (1) At the cost of almost nothing.

Gold, while remaining quite scarce through history, is not immune to increases in supply. If it were ever the case that a new method of mining or acquiring gold became economic, the supply of gold could rise dramatically (cf. 2 Chron. 1:15). Bitcoin is the only commodity whose supply cannot increase when demand increases, compared to gold, where mining capacity can be expanded. Consider the annual production of gold (in blue) and the figure of total bitcoin supply (in yellow).

Established history (trust & adoption): No monetary good has a history as long and storied as gold, which has been valued for as long as human civilisation has existed. Coins minted in the distant days of antiquity still maintain significant value today. The same cannot be said of fiat currencies, which are a relatively recent anomaly of history. From their inception, fiat currencies have had an universal tendency toward eventual worthlessness. The use of “free” money printing as an insidious means of invisibly taxing a citizenry has been a temptation that few states in history have been able to resist. If the 20th century, in which fiat monies came to dominate the global monetary order, established any economic truth, it is that fiat money cannot be trusted to maintain its value over the long or even medium term.

Bitcoin, despite its short existence, has weathered enough trials in the market that there is a high likelihood it will not vanish any time soon. Bitcoin is currently undergoing rapid monetisation, a phenomenon never witnessed before in history for any money. It will likely be several years before Bitcoin transitions from being an incipient store of value to being a medium of exchange.

Censorship-resistance: Bitcoin’s design and engineering meant no single entity (governments, corporations, or individuals) can block, reverse, or arbitrarily restrict valid transactions made with bitcoins. Bitcoin operates on a global, permissionless peer-to-peer network with tens of thousands of nodes and miners securing, protecting, and auditing the network. No central authority controls transaction inclusion. The Bitcoin network is a protocol of rules without rulers. Every node independently verifies all transactions and blocks against Bitcoin’s consensus rules. Invalid transactions are rejected automatically. No one node can force others to accept invalid

data or block valid data. Anyone can run a node, broadcast a transaction, or become a miner without approval. There are no gatekeepers, and no gates to breach.

This property ensures Bitcoin remains a neutral, borderless, decentralised, and peer-to-peer monetary system. Users cannot be deplatformed or prevented from transacting based on political views, geography, beliefs, or any arbitrary criteria or requirements (Rev. 13:17), offering significant financial sovereignty and resistance against authoritarian control and coercion, which is inevitable near the end of this age.

The Recovery of Gold in the Digital Age

For close to 6000 years, gold has been the best form of money used by man for trading, exchanging, transacting, and storing economic energy; as a store of value, as a medium of exchange, and gradually as a unit of account. Like every other metal in the periodic table, it was created by God and given for a special purpose as a precious metal for man to use as a medium that can store and exchange value. In fact, the phrase “sound money” originated from the unique sound pure gold coins make when dropped on hard surfaces. To verify their authenticity and quantity with brevity, gold coins were dropped to measure their “soundness” against another golden coin with a known quantity and quality.

Let us reconsider the monetary properties of gold prior to our modern times, a typical time in most of man’s history, such as when the children of Israel were living in the land of Canaan as detailed in the Chronicles of the Bible.

	Gold	Gold in the 21 st century
Durable	A+	A+
Portable	C	C
Fungible	A	A
Verifiable	A+	C
Divisible	A	A
Scarce	A+	A+
Established history	A+	A+
Censorship resistant	A+	C

Certain monetary properties of gold have drastically diminished in the modern world. As civilisations expanded to the ends of the earth, trading changed from local and communal to

global and long-distance, drastically diminishing its portability. Though technological advancements will never enable man to create gold as was the dreams of the medieval alchemists, it has given man many sophisticated methods to counterfeit gold; examples being under alloying or gold-plated tungsten bullions.



Through the gradual growth and development of civilisation, societies, nation states, and economies toward the 21st century, the economies of scale for gold inevitably led to its centralisation due to its physical attributes. As of today, almost half of the gold ever mined is locked up and stored away in underground bunkers in secrecy, rather than circulating in the economy as money. The creation of modern banking has severed the use of gold as a medium of exchange and centralised its control in the hands of a few banking institutions, which are under the control of their respective governments.

By the 19th century, with the development of modern banking and the improvement in methods of communication, individuals could transact with paper money and checks backed by gold in the treasuries of their banks and central banks. This made gold-backed transactions possible at any size and scale, thus obviating the need for silver's monetary role... Its tragic flaw, however, was that by centralizing the gold behind the iron bars and vaults of banks, and later central banks, it made it possible for banks and governments to increase the supply of money beyond the quantity of gold they held, gradually and implicitly shifting people's perception of value from gold itself to the trust of governments' promise that their issued paper is backed by an equal value of gold.

Since the beginning of the 20th century, the engineering of the modern banking system has gradually and unnaturally demonetised gold (not of its own will) from a decentralised neutral money to a centralised financial asset. A century ago, if you were living in a developed Western nation in North America or Europe and wanted to store gold at a bank, the only institutions at which your gold would still be there after a century were the Bank of England in London, or banks in Switzerland; everywhere else gold was stored, would've been lost due to theft, confiscation (i.e. Executive Order 6102) or war. Because of its vulnerability to centralisation and confiscation, gold's censorship resistance has been significantly diminished.

Had government fiat been a superior store of value across all types of moneys, it would not need government legal tender laws to enforce it, nor would governments worldwide have had to confiscate large quantities of gold and continue to hold them in their central bank reserves. The fact that central banks continue to hold onto their gold, and have even started increasing their gold reserves, testifies to the confidence they have in their own currencies in the long term. (2)

Tragically, even for the best form of money as man's disposal, being the most established, the most durable, the scarcest, and the global form of physical money recognised and accepted universally, it still suffers from:

Being counterfeited
Being centralised
Being confiscated
Being censored

Being not easily divisible
Being not easily verifiable
Being not well portable
Being too unwieldy

With central banks' monopoly position and restrictions on gold ownership and the enforcement of fiat currency by the law across all nations around the globe, never again would gold return to being the world's homogeneous currency. The introduction of Bitcoin, a currency transcending national borders and existing outside the realm of governmental control, offers an intriguing possibility for the emergence of a new type of money. In a way, a recovery of money akin to gold was much needed after the gradual devolution of money since the departure of the gold standard in 1971. This leads us to the question: without changing the intrinsic monetary properties of gold, based on gold's fatal weaknesses, what can we do to recover it? We can make gold better to preserve value as a store of value by:

- Making gold decentralised by giving its users across the globe, on a voluntary basis, a public and digital record and copy of the network.
- Making gold impossible to counterfeit by auditing every existing atom in circulation and every transaction made periodically.
- Making gold censorship resistant by removing all intermediaries, making transacting permissionless and peer-to-peer.

We can make gold easier to transact as a medium of exchange by:

- Making gold weightless.
- Making gold teleportable to the recipient.
- Making gold easily subdivisible at any time when needed.
- Making gold transparent and safer to use.
- Making gold more portable by giving it the ability to be stored in your mind.

This is where Bitcoin steps into the frame. The point of near-perfection is approached when nothing remains to be taken away to improve it. How was Bitcoin engineered? It was engineered such that all the fatal flaws of every type of money were removed, and all the positive attributes were retained. It is akin to refining gold: the more impurities and dross are removed, the more condensed, pure, and valuable the remaining parts become. Bitcoin's monetary purity makes its purpose simple: a vessel that stores and transfers economic energy in a trustless and permissionless way across time and space, in a way that is uncontrollable and unstoppable by any single entity.

The idea of keeping all the upsides and removing all the downsides of gold as money was no novel concept, in fact, this was the idea of Satoshi Nakamoto, the creator of Bitcoin, who portrayed this metaphorical idea of a "digital gold". In a sense, "digital gold" marks the digital transformation of a store of economic energy. A recovery of gold that God intended it to be used as money for man when He permitted the existence of money, lost through the centralisation and restrictions of its ownership enforced by the monetary authorities of this world.

satoshi
Founder
Sr. Member


Activity: 364
Merit: 4756

[Ignore](#)

Re: Bitcoin does NOT violate Mises' Regression Theorem
 August 27, 2010, 05:32:07 PM
Merited by mindrust (2), BlackHatCoiner (2), Gyrsur (1), Husna QA (1), solosequenosenedada (1), RainbowKun (1)
[quote](#) #10

As a thought experiment, imagine there was a base metal as scarce as gold but with the following properties:

- boring grey in colour
- not a good conductor of electricity
- not particularly strong, but not ductile or easily malleable either
- not useful for any practical or ornamental purpose

and one special, magical property:

- can be transported over a communications channel

If it somehow acquired any value at all for whatever reason, then anyone wanting to transfer wealth over a long distance could buy some, transmit it, and have the recipient sell it.

Satoshi's Intent

What exactly was Satoshi Nakamoto trying to accomplish with Bitcoin? For that, we must dive into his writings and correspondences in the early days of the Bitcoin network. He was desperate to provide an alternative not only to financial institutions, but also to fiat currencies prone to devaluation by governments and central banks that replaced gold, an intent evident from his early emails and forum posts. On January 3, 2009, the first Bitcoin block ever mined by Satoshi himself included an embedded message into the genesis block:

The Times 03/Jan/2009 Chancellor on brink of second bailout for banks.

The events of the great financial crisis must have been highly antagonistic to the creator of Bitcoin, for him to place a British newspaper headline about the ongoing financial crisis directly

into the ledger's permanent record. By embedding this cryptic message, he proposed a decentralised and transparent system of money and transactions that offered a necessary evolution and a potential solution to the centralised, enigmatic, and bailout-prone international banking system.

After Bitcoin had been up and running for a few weeks, Satoshi provided a little more detail on his motivations for the project and demonstrated an acute awareness of the instability of credit-money systems, fractionally reserved layers of the fiat money pyramid, and the authority and abuse of fiat by whoever has authority over it:

The root problem with conventional currency is all the trust that's required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust. Banks must be trusted to hold our money and transfer it electronically, but they lend it out in waves of credit bubbles with barely a fraction in reserve.

Satoshi revealed his ambitions for Bitcoin to exist as a currency, not solely as a payments network. Satoshi's criticism of fiat currencies demonstrated a cognisance of the instabilities within our fractionally reserved, layered-money system, under an authoritarian control within the jurisdiction of that fiat's monetary system, regardless of that state's political ideologies. In hindsight, the criticism does appear to have motivated him to create Bitcoin. Perhaps the fabricator wanted to provide the world with a new first-layer money free from the control and corruption of any state, entity, or individual. (3)

